Imperial College London

Name .....

Department of Electrical and Electronic Engineering

Second Year Electronics Laboratory

EEE & EIE Student Handbook

2016-17





Version 5.1

General Information

Students must undertake either 8 (EEE) or 5 (EIE) experiments; apart from PCB, which is not examined, each one has 12 hours of allocated laboratory time. By default, each student will be allocated a partner by the 2nd Year Laboratory Organiser (Prof. Stepan Lucyszyn, s.lucyszyn@imperial.ac.uk) before each experiment. A list will be posted in the teaching laboratory and e-mailed out before the start of each new experiment.

In addition to an Academic Demonstrator, each experiment will usually have two Graduate Teaching Assistants (GTAs) per session. Moreover, the 2nd Year Laboratory Technician (Ms May Tang, m.tang@imperial.ac.uk) will be on hand for general queries.

Please try to make the 2nd Year Laboratory Technician's work easier by **cleaning up your own bench after you have finished your work! In the interests of safety, please report ALL faulty equipment immediately to your academic demonstrator, GTA or any lab technician. Under no circumstances attempt to try to repair equipment; only the lab technicians should do this!**

For each experiment, a Lead Academic is responsible for particular experiments and they have prepared a set of handouts for the students; 2nd Year Laboratory Technician will issue a hard copy to each student, along with a new logbook (one for each experiment). For more information, relevant lab web pages can be found on Blackboard.

If you have any technical questions relating to a particular experiment, and you cannot find the answers on the handouts or intranet web pages, please contact the Lead GTA in the first instance; failing that, contact the Lead Academic directly assigned to that experiment. For all other queries relating the 2nd Year Labs, please do not hesitate to contact the 2nd Year Lab Organiser.

Attendance

Laboratory work is really important, as it should enhance your understanding of taught courses in the classroom and so regular attendance is expected. A register of attendance is maintained via the RFID *touch-n-go* reader; absences from the scheduled lab sessions will be periodically reviewed. Anyone who regularly misses their lab sessions will be reported to the Senior Tutor (Mrs Esther Perea, e.perea@imperial.ac.uk) and your Personal Tutor; a record of this will also be placed in your personal file (this will be seen by those that have the task of writing references on your behalf and also Fact Finders that present evidence to the examiners' meetings).

If you know you are going to be absent for a particular session, ahead of time, please inform the 2nd Year Lab Organiser. If you are absent for more than one session, perhaps for health/personal reasons, then you must inform the Senior Tutor directly; Mrs Perea will help to determine if any support needs to be put in place or whether you should consider submitting a Mitigating Circumstances form.

Under normal circumstances, there are NO scheduled catch-up sessions. You must complete your experiment with the allocated 12 hours. However, if you are unable to attend your scheduled lab session then you are encouraged to use the lab outside of scheduled times, BUT you need to submit a valid reason to the 2nd Year Lab Organiser in writing (not being able to get out of bed, having a terrible hangover or disappearing for long weekends, etc., are NOT valid reasons). For medical reasons, evidence must be sent to the Senior Tutor directly. You will then have to arrange for a mutually suitable time slot with the 2nd Year Lab Technician.

Laboratory Partners

For the 2nd year labs, students will be pseudo-randomly allocated a new lab partner (where possible) before each and every experiment; you cannot choose your own lab partners! There are a number of important reasons for our allocation process; these include the following:

1. In the real world it is very likely that you will have to work closely with a spectrum of people (in terms of ethnicity, nationality, race, heritage, culture, gender ... and personality). To prepare you for this real-life experience, you may be allocated up to 8 (EEE) or 5 (EIE) different lab partners.
2. Since you will only have to spend a maximum of 12 hours with each partner, you will not be 'stuck' with someone for the whole academic year.
3. It is easier to report any imbalances of effort when it is not your best friend!
4. Helps to avoid any undue bias in the marking of experiments by the GTAs.
5. Raises the overall standards with both the learning environment and assessment process.
6. Most students actually prefer this.

If you are allocated a partner that you are unwilling to work with then you must e-mail the 2nd Year Lab Organiser (and copy to the Senior Tutor) before the experiment; clearly explaining your objections. This e-mail may be kept on record.

Note that, while you have been allocated lab partners, you are expected to be self-reliant on undertaking all parts of the experiments – at all times!

For logistical reasons, a very few randomly selected students will not be allocated a partner. This should be considered an advantage, since they can work without distractions and at their own pace.

Logbooks

In the real world it may be necessary to take a detailed look back at previous work – perhaps long after you remember even doing the experiment. For example, if a patent is being challenged, the future success of your company or institution may depend on whether you are able to convince a court of law that you actually worked on a particular invention and achieved certain results. A logbook provides real evidence of your work and contains the raw data you collected, but this is not the same as a report. Unlike reports, which are carefully written after the work has been undertaken, logbooks must be a detailed record of your work taken at the time you did the work!

Traditional practice dictates that the logbook should be a bound or stapled notebook, named and dated on the first page, and its pages should be numbered to make it impossible to tear off pages without being obvious. Though paper logbooks are now often replaced with digital logbooks, such as source control or content management systems, these serve the same purpose of providing timestamped and sequential evidence of your ideas and data. In order to focus on the basic principle of documenting and logging your activities and thoughts, we still require a paper-based logbook.

Use your logbook for drawing your designs, performing calculations, recording important equipment and their settings, writing experimental results, noting both raw and processed data and making observations and conclusions.

Do not waste time 'writing-up' your logbook after the lab session, since the logbook must be a contemporaneous note. **See also the note about plagiarism.** It is good practice to learn to keep reasonably neat and understandable notes in a logbook during the lab session, as you carry out the experiment. Any computer printouts should be pasted into the logbook. If you make errors, it is better to cross it out, and write a (dated!) reference pointing to the page where the corrected version exists. Correcting fluid may

appear neat, but can also lead to a lot of confusion when multiple corrections accumulate. See the assessment guidelines on the lab web pages for further instructions on how to write logbooks.
Finally, you must submit your logbook to the 2nd Year Laboratory Technician at the end of the allocated 12 hours! A GTA will look through your logbook and ask questions from it during your short interview. The logbook can be obtained from the 2nd Year Laboratory Technician, once the assessment procedure is complete for that particular experiment.

Experiments

The lab experiments are designed to complement taught courses in the 2nd year. However, due to logistical reasons, it is not always possible to schedule them so that experiments follow on from what has been taught. In addition, since some experiments require more effort than others, we have tried to spread-out the workload across the year.

Exp. Ref. Number	Acronym	Short Description	Related Course	Lead Academic Responsible
AUTUMN TERM				
1	DFT	Linear Signal Processing in MATLAB	Signals & Linear Systems	Dr Bruno Clerckx b.clerckx@imperial.ac.uk
2	AMP	High Gain Amplifier, **requiring a demonstration**	Analogue Electronics	Dr Christos Papavassiliou c.papavas@imperial.ac.uk
3	VERI	Digital Design with Verilog	Digital Electronics	Prof. Peter Y. K. Cheung p.cheung@imperial.ac.uk
SPRING TERM				
4	PCB	Printed Circuit Board, **needed for the SWITCH experiment**	-	Dr Edward Stott ed.stott@imperial.ac.uk
5	COMMS	Communications Modulation & Demodulation Systems	Communication Systems	Dr Deniz Gunduz d.gunduz@imperial.ac.uk
6	SMPS	Switch-Mode Power Supply	Power Engineering	Dr Balarko Chaudhuri b.chaudhuri@imperial.ac.uk
7b	BULB	Light Bulb Control	Control Engineering	Dr Imad Jaimoukha i.jaimouka@imperial.ac.uk
or				
7m	MACH	Field Plotting and Induction Machines	Fields and Power Engineering	Prof. Bikash Pal b.pal@imperial.ac.uk
SUMMER TERM				
8	SWITCH	Touch Switch, **requiring a demonstration**	Computer Architecture	Dr Tom Clarke t.clarke@imperial.ac.uk

Schedule for EEE Students Only

(All session from 9 am to 12 noon in the Teaching Laboratory, Room 114B)

Autumn Term 2016									
Week	Monday		Tuesday		Thursday		Friday		
	Ax	Ay	Bx	By	Ax	Ay	Bx	By	
1	FREE								
2-5	DFT	AMP	DFT	AMP	AMP	DFT	AMP	DFT	
6	FREE (Interviews take place during this week)								
7-10	VERI		VERI			VERI		VERI	← E copy Labsheet in logbook
11	FREE (Interviews take place during this week)								

Spring Term 2017									
Week	Monday		Tuesday		Thursday		Friday		
	Bx	By	Ax	Ay	Bx	By	Ax	Ay	
15	PCB	PCB	PCB	PCB	PCB	PCB	PCB	PCB	
16-19	COMMS	SMPS	COMMS	SMPS	SMPS	COMMS	SMPS	COMMS	← e-logbook
20	FREE (Interviews take place during this week)								
21-24		* BULB/ MACH		* BULB/ MACH	* BULB/ MACH		* BULB/ MACH		
25	FREE (Interviews take place during this week)								

Summer Term 2017											
Week	Monday		Tuesday		Wednesday		Thursday		Friday		
	Ax	Ay	Bx	By	Ax	Ay	Ax	Ay	Bx	By	
31	Bank Holiday		SWITCH	SWITCH	SWITCH	SWITCH	SWITCH	SWITCH	SWITCH	SWITCH	← E-Copy of labsheet
32	SWITCH	SWITCH	SWITCH	SWITCH			SWITCH	SWITCH	SWITCH	SWITCH	
33-39	FREE										

* Lab pairs will do either the BULB or MACH experiments. Individual students must register their preference (if they have one) by the end of week 19 and every effort will be made to accommodate.

Schedule for EIE Students Only

(All session from 9 am to 12 noon in the Teaching Laboratory, Room 114B)

Autumn Term 2016									
Week	Monday		Tuesday		Thursday		Friday		
			Ix					Iy	
1	FREE								
2-5			DFT					DFT	
6	FREE (Interviews take place during this week)								
7-10			VERI					VERI	
11	FREE (Interviews take place during this week)								

Spring Term 2017									
Week	Monday		Tuesday		Thursday		Friday		
	Ix	Iy			Ix	Iy			
15	PCB	PCB			PCB	PCB			
16-19	COMMS					COMMS			
20	FREE (Interviews take place during this week)								
21-24									
25	FREE (Interviews take place during this week)								

Summer Term 2017										
Week	Monday		Tuesday		Wednesday		Thursday		Friday	
			Ix	Iy					Ix	Iy
31			SWITCH	SWITCH					SWITCH	SWITCH
32			SWITCH	SWITCH					SWITCH	SWITCH
33-39	FREE									

Academic Demonstrators

(* Lead academic for the associated experiment)

Day	Experiments (Streams)	Academic	Email Address @imperial.ac.uk
AUTUMN TERM 2016 (Weeks 2-5, 7-10)			
Monday	**DFT, AMP** (Ax, Ay) Weeks 2-5	Dr Simos Evangelou, Dr Christos Papavassiliou *	s.evangelou c.papavas
	VERI (Ax) Weeks 7-10	Prof. Peter Cheung *	p.cheung
Tuesday	**DFT, AMP** (Bx + Ix, By) Weeks 2-5	Dr Bruno Clerckx *, Prof. Stepan Lucyszyn	b.clerckx s.lucyszyn
	VERI (Bx + Ix) Weeks 7-10	Dr Edward Stott	ed.stott
Thursday	**DFT, AMP** (Ay, Ax) Weeks 2-5	Dr Simos Evangelou, Dr Edward Stott	s.evangelou ed.stott
	VERI (Ay) Weeks 7-10	Mike Brookes	mike.brookes
Friday	**DFT, AMP** (By + Iy, Bx) Weeks 2-5	Dr Bruno Clerckx *, Dr Christos Papavassiliou *	b.clerckx c.papavas
	VERI (By + Iy) Weeks 7-10	Dr Edward Stott	ed.stott
SPRING TERM 2017 (Weeks 15, 16-19, 21-24)			
Monday	**PCB** (Bx + Ix, By + Iy) Week 15	Dr Tom Clarke, Dr Edward Stott *	t.clarke ed.stott
	COMMS, SMPS (Bx + Ix, By) Weeks 16-19	Dr Deniz Gunduz *, Dr Balarko Chaudhuri *	d.gunduz b.chaudhuri
	BULB/MACH (By) Weeks 21-24	Dr Tom Clarke	t.clarke
Tuesday	**PCB** (Ax, Ay) Week 15	Dr Tom Clarke, Dr Edward Stott *	t.clarke ed.stott
	COMMS, SMPS (Ax, Ay) Weeks 16-19	Dr Tom Clarke, Prof. Bikash Pal	t.clarke b.pal
	BULB/MACH (Ay) Weeks 21-24	Dr Eric Kerrigan	e.kerrigan
Thursday	**PCB** (Bx + Ix, By + Iy) Week 15	Dr Tom Clarke, Dr Edward Stott *	t.clarke ed.stott
	COMMS, SMPS (By + Iy, Bx) Weeks 16-19	Dr Deniz Gunduz *, Dr Balarko Chaudhuri *	d.gunduz b.chaudhuri
	BULB/MACH (Bx) Weeks 21-24	Patricio Petruzzi	p.petruzzi12
Friday	**PCB** (Ax, Ay) Week 15	Dr Tom Clarke, Dr Edward Stott *	t.clarke ed.stott
	COMMS, SMPS (Ay, Ax) Weeks 16-19	Dr Christos Papavassiliou, Prof. Bikash Pal	c.papavas b.pal
	BULB/MACH (Ax) Weeks 21-24	Dr Edward Stott	ed.stott
SUMMER TERM 2017 (Weeks 31-32)			
Monday	**SWITCH** (Ax, Ay) Week 32	Dr Tom Clarke * Dr Edward Stott	t.clarke ed.stott
Tuesday	**SWITCH** (Bx + Ix, By + Iy) Weeks 31-32	Dr Tom Clarke * Dr Edward Stott	t.clarke ed.stott
Wednesday	**SWITCH** (Ax, Ay) Week 31	Dr Tom Clarke * Dr Edward Stott	t.clarke ed.stott
Thursday	**SWITCH** (Ax, Ay) Weeks 31-32	Dr Tom Clarke * Dr Edward Stott	t.clarke ed.stott
Friday	**SWITCH** (Bx + Ix, By + Iy) Weeks 31-32	Dr Tom Clarke * Dr Edward Stott	t.clarke ed.stott

Your Graduate Teaching Assistants (GTAs)

(* Lead GTA for the associated experiment)

Acronym	Days	GTA Names	GTA Email Addresses @imperial.ac.uk
DFT			
AMP			
VERI			

Acronym	Days	GTA Names	GTA Email Addresses @imperial.ac.uk
PCB			
COMMS			
SMPS			
BULB/MACH			

Acronym	Days	GTA Names	GTA Email Addresses @imperial.ac.uk
SWITCH			

Assessment

As with any other forms of assessment, the Department is looking to differentiate between weak and strong students; for the 2nd year labs the mean letter grade for any experiment and overall should be B.

All experiments have equal marks.

You must submit your logbook by placing it in the corresponding labelled container provided by the 2nd Year Laboratory Technician, located in Room 114B, by 12 noon immediately after the very last scheduled session for your stream:

Acronym	Week	Deadline Dates
AUTUMN TERM		
DFT	5	31st Oct. – 4th Nov. 2016
AMP (EEE only)	5	31st Oct. – 4th Nov. 2016
VERI	10	5th – 9th Dec. 2016
SPRING TERM		
PCB	15	12th – 13th Jan. 2017
COMMS	19	6th – 10th Feb. 2017
SMPS (EEE only)	19	6th – 10th Feb. 2017
BULB/MACH (EEE only)	24	13th – 17th Mar. 2017
SUMMER TERM		
SWITCH	32	11th – 12th May 2017

No marks will be allocated for the neatness of the logbook, but marks will be deducted if the GTA cannot read your writing!

Your logbook will be brought to the interview by the randomly assigned GTA, so that the student can use it to answer questions about the experiment. The 15-20 minute interview will be with an individual student or lab pair. The GTA will ask questions relevant to that particular experiment and not just referring to you logbook. NOTE that while a lab pair may have a shared interview, separate marks will be allocated to each student.

The student's logbook has a weighting of 40%, while the interview has a weighting of 60%.

All marks are moderation, to remove bias between GTAs. An individual's moderated marks will then be recorded, based on understanding of the experiment, achievements in obtaining the correct results, observations and conclusions reached. A letter grade will appear on Blackboard and written feedback sent by e-mailed to each student within approximately two weeks of your interview.

Please note that a 1% error in the marking of any 2nd year lab assessment will result in a worst-case error of between 0.002% and 0.006% in your overall degree. However, having a total of 14(8) assessments for the EEE(EIE) students, on average, you are equally likely to have been over-marked as under-marked and so there is an inherent moderation mechanism.

If you believe that you have been unfairly graded then please e-mail the 2nd Year Lab Organiser, stating the nature of your concern (citing evidence, as opposed to hearsay). If appropriate, your case will be passed on to the Lead Academic responsible for the experiment and associated GTAs.

If, after an investigation by the Lead Academic and GTAs, it has been found that you have not been unfairly marked then you will not be allowed to challenge your grade from subsequent experiments.

Interview Marking Boundary Definitions

Grade A* (80% or above): The student shows an outstanding grasp of the experiment or design trade-offs, and/or has made an extraordinary and original contribution to the design.

Grade A (70% to 79%): The student shows an excellent understanding of the theory and practice of the experiment or design. Demonstrated work is completely successful, and some aspects of the students work or understanding go beyond what is required and/or demonstrate unusually deep understanding or flair for or creativity in the design.

Grade B (60% to 69%): The student shows a good understanding of the theory and practice of the experiment or design. Demonstrated work is largely operational, and the remaining problems have been correctly detected by the student.

Grade C (50% to 59%): The student shows a basic understanding of the theory and practice of the experiment or design. Significant parts of the demonstrated work function correctly.

Grade D (40% to 49%): The student shows a limited understanding of some aspects of the theory and/or practice of the experiment or design. The demonstrated work has some elements that work correctly.

Grade E (below 40%): The student shows almost no understanding of the experiment or design and the demonstrated work is of little value.

WARNINGS

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STUDENTS FOUND COPYING MATERIAL FROM ANYONE (OTHER THAN SHARED RAW DESIGN/EXPERIMENTAL DATA WITH THEIR ASSIGNED LAB PARTNER) WILL RECEIVE A ZERO MARK FOR THE EXPERIMENT

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College regulations state that late coursework submissions should receive zero marks unless there are mitigating circumstances. For medical reasons, evidence must be sent to the Senior Tutor.

In addition, late logbooks will not be given feedback. Exceptions to this rule are given in the Departmental documentation, and cannot be granted by the 2nd Year Lab Organiser.

Missing a scheduled interview time slot* will result in a zero mark, unless the Departmental guidelines for mitigating circumstances are met, in which case an interview will need to be arranged between the student and a GTA, after permission has been granted by the Senior Tutor.

*** In this case only your partner will be interviewed during the scheduled time slot.**